Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.05	20.25	
worker nodes in service, mean	6.00	5.05	-15.8%
worker nodes in service, min	6.00	5.00	-16.7%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.00	1.70	-15.0%
power (W), mean	624	566	-9.2%
power (W), peak	646	620	-4.2%
energy (Wh)	208	191	-8.3%
energy, parked workers still on at idle power (Wh)	208	215	+3.2%
CPU used (cores), mean	0.751	1.10	+46.5%
CPU allocatable (cores), mean	24.00	20.20	-15.8%
utilisation (used / allocatable)	0.031	0.054	+74.1%
energy per core-hour (Wh)	831	515	-38.0%
node-hours per core-hour	7.99	4.59	-42.5%
pending pods, pod-minutes	0.267	0.000	-100.0%
pending pods, peak	1.00	0.000	-100.0%
HPA replicas, mean	8.83	9.28	+5.1%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.533	0.800	+50.0%
requests timed	1,135	1,120	-1.3%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	57.95	75.37	+30.1%
response time (ms), median	40.80	68.00	+66.7%
response time (ms), 95th percentile	129	105	-19.0%
response time (ms), 99th percentile	207	134	-35.1%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	8.9789	5.2060	-42.0%	[-5.8684, -1.6541]	GAIN
kwh_per_core_hour	0.9290	0.5780	-37.8%	[-0.5581, -0.1467]	GAIN
utilisation	0.0310	0.0541	+74.4%	[+0.0100, +0.0372]	GAIN
pending_pod_minutes_per_hour	0.7742	0.0000	-100.0%	[-2.3226, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	1.5610	2.3227	+48.8%	[-3.1345, +3.8774]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 1.0 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.359 0.392 0.417 0.438 0.457 0.473 0.488 0.5 0.512 0.522 0.531 0.536 0.54 0.544 0.548 0.552 0.555 0.557 0.559 0.56
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	614	618	5.00	5.00	0.000	0.000	0.473	0.572

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	5.00	612	549	5.00	6.00	0.000	0.000	0.436	0.731
2	6.00	5.00	612	551	5.00	7.50	0.000	0.000	0.476	0.786
3	6.00	5.00	616	555	5.00	8.00	0.000	0.000	0.552	0.873
4	6.00	5.00	630	571	7.25	9.50	0.000	0.000	0.905	1.30
5	6.00	5.00	628	572	10.00	10.00	0.000	0.000	0.845	1.36
6	6.00	5.00	628	568	10.00	10.00	0.250	0.000	0.869	1.26
7	6.00	5.00	644	585	10.00	10.00	0.000	0.000	1.29	1.68
8	6.00	5.00	640	588	10.00	10.00	0.000	0.000	1.16	1.80
9	6.00	5.00	642	593	10.00	10.00	0.000	0.000	1.27	1.91
10	6.00	5.00	636	578	10.00	10.00	0.000	0.000	1.10	1.52
11	6.00	5.00	612	547	10.00	10.00	0.000	0.000	0.456	0.693
12	6.00	5.00	613	546	10.00	10.00	0.000	0.000	0.466	0.643
13	6.00	5.00	616	554	10.00	10.00	0.000	0.000	0.540	0.860
14	6.00	5.00	632	569	10.00	10.00	0.000	0.000	0.947	1.27
15	6.00	5.00	628	564	10.00	10.00	0.000	0.000	0.841	1.16
16	6.00	5.00	630	567	10.00	10.00	0.000	0.000	0.914	1.23
17	6.00	5.00	617	558	10.00	10.00	0.000	0.000	0.571	0.966
18	6.00	5.00	612	547	10.00	10.00	0.000	0.000	0.448	0.686
19	6.00	5.00	614	546	10.00	10.00	0.000	0.000	0.486	0.659

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.